

Intro to Syllabus Class Commands Usage

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Structure of a Latex document

```
\documentclass[] {syllabus}
```

PREAMBLE Section

```
\begin{document}
```

BODY Section

```
\end{document}
```

- Packages: Include any additional packages you need to enhance the functionality of LaTeX. Packages extend LaTeX's capabilities and provide features for specialized purposes, such as graphics, tables, mathematical symbols, and citation styles.
- Title, author, and date: Set the title of your document, the author's name, and the date. This information is typically used to generate a title page or header/footer in the document.
- ...

Syllabus template [v1.0]

How to access "Syllabus template" zip file

Head to the following website:

<https://github.com/rvce-latex/syllabus>

Installation

Introduction

- Utilizes standard open-source tools:

Compiler



Miktex

<https://miktex.org/download>

Latex Editor



TexStudio

<https://www.texstudio.org/>

For Windows: **Miktex** compiler.

For MacOS: **Mactex** compiler.

For Linux: **TeX Live** compiler.

- **Miktex:** It's a like C compiler
<https://miktex.org/download>
- **TeXstudio/TeXmaker:** It's a code editor, which provides more functionality than the native editor(TeXworks) that ships with Miktex <https://www.texstudio.org/>

- **JabRef:** It's a free reference manager
<https://www.jabref.org/download/win>
- **Strawberry Perl:** Tool needed to properly compile your Bibliography
<https://strawberryperl.com/>

Introduction

Introduction to Syllabus class

- Collects and typeset course syllabus information in a structured way.
- Defines consistent interface for specific programme options, departments, semesters, and scheme.
- Intended for faculty and curriculum committees to generate syllabi documents in a standardized layout with minimal manual formatting.

Introduction to Syllabus class



Go, change the world

Semester - III		
Applied Design thinking lab		
Category: Professional Core Course		
(Common to Courses in CSE/IT/CE and EE Programs)		
(Practical)		
Course Code	1	CSE304L
Credits L/T/P	1	0/0/1
Total Hours	1	2/0/0
Hardware requirements:		
How you can get your instructions here: If you don't want it:		
1. Design		
Innovative Experiments		
2. RAI for EE experiment		
Course Outcomes: After completing the course, the:		
1. sample		
References		
[1] B. Ramani, Design of Analog CMOS Integrated Circuits, 0072180222		

BUBBLES FOR THE CONTINUOUS INTERNAL E	
#	COMPONENTS
1	QUESTIONS: Questions will be conducted in multiple choice mode. QUESTIONS will be conducted and each question will be THE SUM OF TWO WILL BE THE FINAL MARKS.
2	TESTS: Students will be evaluated in test, at different complexity levels (Based on Revised 1) and THREE tests will be conducted.
3	EXPERIMENTAL LEARNING: Students will conduct and practical implementation of the pre teaching learning (PL). Program specific requirements (Implementation) (IS). Task: UPTO 40 MARKS.
MAXIMUM MARKS	



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Semester - III		
Environment and Sustainability		
Category: Professional Core Course		
(Common to Courses in CSE/IT/CE and EE Programs)		
(Theory)		
Course Code	1	CV202A
Credits L/T/P	1	0/0/1
Total Hours	1	0/0/1
Unit - I		
sample		
Course Outcomes: After completing the course, the:		
1. sample		
References		
[1] B. Ramani, Design of Analog CMOS Integrated Circuits, 0072180222		

BUBBLES FOR THE CONTINUOUS INTERNAL E	
#	COMPONENTS
1	QUESTIONS: Questions will be conducted in multiple choice mode. QUESTIONS will be conducted and each question will be THE SUM OF TWO WILL BE THE FINAL MARKS.
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MAXIMUM MARKS	

BUBBLES FOR THE CONTINUOUS INTERNAL E	
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MAXIMUM MARKS	

BUBBLES FOR SEMESTER END EXAMINATION	
Q.No.	CONTENTS
1	Objective type questions covering entire syllabus



Go, change the world

Semester - III		
Linear Algebra and Probability Theory		
Category: Professional Core Elective		
(Common to Courses in CE and EE)		
(Theory & Statistical)		
Course Code	1	MA202U
Credits L/T/P	1	0/1/0
Total Hours	1	0/1/0
Unit - I		
sample		
Course Outcomes: After completing the course, the:		
1. sample		
References		
[1] B. Ramani, Design of Analog CMOS Integrated Circuits, 0072180222		

BUBBLES FOR THE CONTINUOUS INTERNAL E	
#	COMPONENTS
1	QUESTIONS: Questions will be conducted in multiple choice mode. QUESTIONS will be conducted and each question will be THE SUM OF TWO WILL BE THE FINAL MARKS.
2	TESTS: Students will be evaluated in test, at different complexity levels (Based on Revised 1) and THREE tests will be conducted.
3	EXPERIMENTAL LEARNING: Students will conduct and practical implementation of the pre teaching learning (PL). Program specific requirements (Implementation) (IS). Task: UPTO 40 MARKS.
MAXIMUM MARKS	

BUBBLES FOR THE CONTINUOUS INTERNAL E	
#	COMPONENTS
1	QUESTIONS: Questions will be conducted in multiple choice mode. QUESTIONS will be conducted and each question will be THE SUM OF TWO WILL BE THE FINAL MARKS.
2	TESTS: Students will be evaluated in test, at different complexity levels (Based on Revised 1) and THREE tests will be conducted.
3	EXPERIMENTAL LEARNING: Students will conduct and practical implementation of the pre teaching learning (PL). Program specific requirements (Implementation) (IS). Task: UPTO 40 MARKS.
MAXIMUM MARKS	

BUBBLES FOR SEMESTER END EXAMINATION	
Q.No.	CONTENTS
1	Objective type questions covering entire syllabus



Go, change the world

Semester - III		
Data Structure and Algorithms		
Category: Professional Core Course		
(Common to Courses in CE and EE)		
(Theory & Practical)		
Course Code	1	CV202A
Credits L/T/P	1	0/0/1
Total Hours	1	0/0/1
Unit - I		
sample		
Course Outcomes: After completing the course, the:		
1. sample		
References		
[1] B. Ramani, Design of Analog CMOS Integrated Circuits, McGraw-Hill Education, 2005, ISBN: 007-218022-2		

BUBBLES FOR THE CONTINUOUS INTERNAL EVALUATION	
#	COMPONENTS
1	QUESTIONS: Questions will be conducted in multiple choice mode. TWO QUESTIONS will be conducted and each question will be evaluated for 10 Marks. THE AVERAGE OF THE TWO WILL BE THE FINAL Q&E MARKS.
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised 1) and THREE tests will be conducted.
3	EXPERIMENTAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. One study based teaching learning (IS). Program specific requirements (IS). Video based module (practical/implementation) (IS). Task: UPTO 40 MARKS.
MAXIMUM MARKS	

BUBBLES FOR THE CONTINUOUS INTERNAL EVALUATION	
#	COMPONENTS
1	QUESTIONS: Questions will be conducted in multiple choice mode. TWO QUESTIONS will be conducted and each question will be evaluated for 10 Marks. THE AVERAGE OF THE TWO WILL BE THE FINAL Q&E MARKS.
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised 1) and THREE tests will be conducted.
3	EXPERIMENTAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. One study based teaching learning (IS). Program specific requirements (IS). Video based module (practical/implementation) (IS). Task: UPTO 40 MARKS.
MAXIMUM MARKS	

PO course type

LO course type

LT course type

LP course type

Workflow of Syllabus class

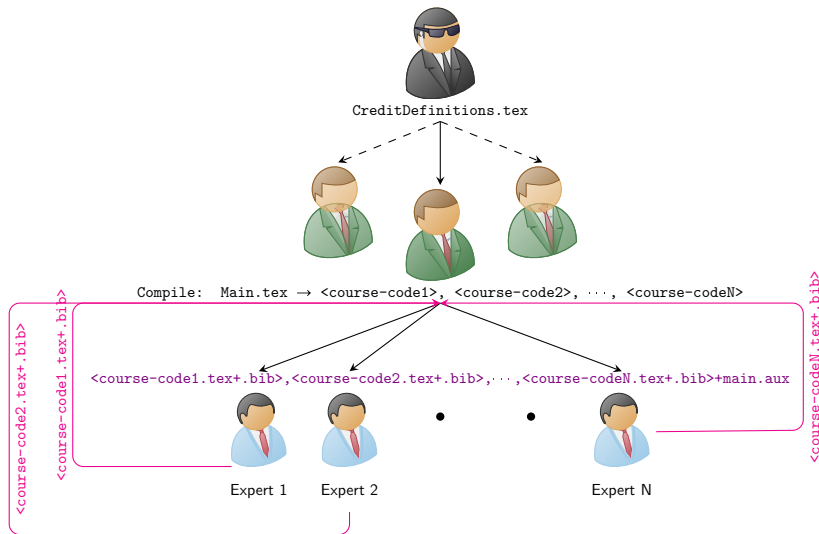


Figure: Workflow of Syllabus.cls from Dean of Academics to HOD and subject experts

Syllabus class

Syllabus class command classification

There are three Major command classifications:

1. Credit-related commands: Used by DA
 - Defines variants to the course **base** type
 - Defines LTP credit values, SEE/CIE duration+Max Marks, rubrics for CIE/SEE with marks distribution for each course-types (defined above).
2. Scheme-related commands: Used by HOD
 - Defines the course-code, course-title, category, common to, bos.
 - Assigns the appropriate course-types (defined by DA) and categories to the courses.
 - Group subject cores according to the following Category: PE, AE and PC.
 - Generate individual <course-code.tex/.bib> file from Main.tex and send it to subject expert.
3. Course-related commands: Used by Subject Experts.
 - Based on the course-type, define the content of units, practicals, course outcomes in <course-code.tex> file.
 - Add references using <course-code.bib> file.

Credit-related commands: Course-type definitions



Dean of Academics (DA)

- Defines the course base types (and it's variants whenever it is required).
- Defines the L:T:P credit distributions to each course types.
- Defines the CIE/SEE Duration/Marks to each course types.
- Defines the CIE/SEE Rubrics contents with marks distributions to each course types.

Credit-related commands: Course-type definitions

```
\begin{definecredits}◀
\hrspercred{hrs/cred}◀ Defines number of hours per credit (in total credits)
\wkspercred{weeks/cred}◀ Defines the number of weeks
\Lcredmul{L cred multiplier}
\Tcredmul{T cred multiplier} }◀ Serves as a multiplier for hours per credit per week
\Pcredmul{P cred multiplier}
\begin{credit}{course base type}◀ Define one among the 4 base course-types: LO, LP, LT, PO
\variantchar{char}◀ Supports polymorphism to create variants in each base course-types.
\Lcredit{number}◀ Defines different credit combinations including variants under each base course-types.
\Tcredit{number}
\Pcredit{number}
% Durations are specified in hours (as numbers w/o any post fix)
\CIEduration{in hours} }◀ Supports different Max. duration distribution: Under each course-types
\SEEduration{in hours}
% Total final Marks of the course
\CIELmarks{number}
\CIEPmarks{number} }◀ Supports different Max. marks distribution: Under each course-types
\SEELmarks{number}
\SEEPmarks{number}
% Draws CIE rubric table at the bottom
\begin{CIErubrics}◀
\Crubric{qno}{rubric text}{marks}◀ Meant to define the rubrics for CIE in the order: <no> <content> <mark>
\end{CIErubrics}
% Draws SEE rubric table at the bottom
\begin{SEE@Theory@rubrics}[duration]◀
\SrubicA{qno}{contents}{marks}◀ Meant to define the Part A rubric for SEE Theory.
\SrubicB{qno}{contents}{marks}◀ Meant to define the Part B rubrics for SEE Theory.
\end{SEE@Theory@rubrics}
\begin{SEE@Practical@rubrics}◀ Meant to define the rubrics for SEE Practical.
\Srubic{qno}{contents}{marks}◀
\end{SEE@Practical@rubrics}
\end{credit}◀
\end{definecredits}◀
```

Credit-related commands: Example

```
\begin{definecredits}
\begin{credit}{LP} ← base course type
% \variantchar{a}
\LCredit{3}
\TCredit{4}
\PCredit{1}
\CIEduration{2}
\SEEduration{3}
\CIELmarks{100}
\CIEPmarks{50}
\SEELmarks{100}
\SEEPmarks{50}
\begin{CIERubrics}
\Crubric{1}{\textbf{QUIZ...}{10}
\Crubric{2}{\textbf{TEST...}{40}
\end{CIERubrics}
\begin{SEE@Theory@rubrics}
\SRubricA{1}{Objective ty...}{20}
\SRubricB{2}{Unit - I: Co...}{16}
\SRubricB{3 & 4}{Unit - II...}{16}
...
\end{SEE@Theory@rubrics}
\begin{SEE@Practical@rubrics}
\SRubric{1}{Write-up}{10}
\SRubric{2}{Conduction}{20}
\SRubric{3}{Viva-voce}{20}
\end{SEE@Practical@rubrics}
\end{credit}
\begin{credit}{LT} ← base course type
% \variantchar{b}
\CIEduration{2}
\SEEduration{3}
\LCredit{2}
```

```
\begin{definecredits}
\begin{credit}{LO}
\variantchar{a} ← variant: LOa
\LCredit{4}
\TCredit{4}
\PCredit{4}
\CIEduration{2}
\SEEduration{3}
\CIELmarks{100}
\SEELmarks{100}
\begin{CIERubrics}
\Crubric{1}{\textbf{QUIZ...}{10}
\Crubric{2}{\textbf{TEST...}{40}
\end{CIERubrics}
\begin{SEE@Theory@rubrics}
\SRubricA{1}{Objective ty...}{20}
\SRubricB{2}{Unit - I: Co...}{16}
\SRubricB{3 & 4}{Unit - II...}{16}
...
\end{SEE@Theory@rubrics}
\end{credit}
\begin{credit}{LOb} ← variant: LOb
\variantchar{b}
\CIEduration{1}
\SEEduration{2}
\LCredit{1}
\CIELmarks{50}
\SEELmarks{50}
\begin{CIERubrics}
\Crubric{1}{\textbf{QUIZ...}{10}
\Crubric{2}{\textbf{TEST...}{20}
\end{CIERubrics}
\begin{SEE@Theory@rubrics}
```

Credit-related commands: Example of LP course base type

```
\begin{definecredits}
\begin{credit}{LP}◀
% \variantchar{a}
\Lcredit{3}
\Tcredit{4}
\Pcredit{1}
\CIEduration{2}
\SEEduration{3}
\CIELmarks{100}◀
\CIEPmarks{50}◀
\SEELmarks{100}◀
\SEEPmarks{50}◀
...
\end{credit}
\end{credit}
\end{definecredits}
```

III SEMESTER														
Sl. No.	Course Code	Course Title	BoS	Credit Allocation			Category	CIE Duration (Hrs)	Max Marks CIE		SEE Duration (Hrs)	Max Marks SEE		
				L	T	P			Theory	Lab		Theory	Lab	
1	MA231TC	Linear Algebra and Probability Theory	MA	3	0	0	0 ¹	Theory	3	100	—	100	—	
2	EA131AIX / EB131AIX	Elements of electronics / Elements of electronics	ECE / ECE	3	0	0	3	Theory	3	100	—	3	100	—
3	EC132AXX	AE Course	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
4	EC134EXX	Advanced Electronics Group	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
5	EC131AIX	Elements of electronics	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
Total				15										

Semester - IV					
Design and Analysis of Algorithms					
Category: Professional Core Course					
(Common to CS,CD,CY and CI Programs)					
(Theory & Practical)					
Course Code	:	CD343AI		CIE	: 100 + 50 Marks
Credit distribution: L:T:P	:	3:0:1		SEE	: 100 + 50 Marks
Pedagogical Hours (L:T:P:S)	:	45:0:30:45		CIE Duration	: 1.5 Hours
Total credits:	:	4		SEE Duration	: 3 Hours

Credit-related commands: Example of LP course base type

```
\begin{definecredits}
\begin{credit}{LP}◀
...
\begin{CIErubrics}◀
\Crubric{1}{\textbf{QUIZ...}}{10}
\Crubric{2}{\textbf{TEST...}}{30}
\Crubric{3}{\textbf{EXPE...}}{30}
\Crubric{4}{\textbf{LAB:...}}{30}
\end{CIErubrics}
\begin{SEE@Theory@rubrics}◀
\SrubicA{1}{Objective ty...}{10}
\SrubicB{2}{Unit - I: Co...}{18}
\SrubicB{3 & 4}{Unit - II...}{18}
\SrubicB{5 & 6}{Unit - II...}{18}
\SrubicB{7 & 8}{Unit - IV...}{18}
\SrubicB{9 & 10}{Unit - V...}{18}
\end{SEE@Theory@rubrics}
\begin{SEE@Practical@rubrics}◀
\Srubic{1}{Write-up}{10}
\Srubic{2}{Conduction}{20}
\Srubic{3}{Viva-voce}{20}
\end{SEE@Practical@rubrics}
\end{credit}
\end{credit}
\end{definecredits}
```

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE AVERAGE OF THE TWO WILL BE THE FINAL QUIZ MARKS.	10
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	30
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 30 MARKS.	30
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (30 Marks), Lab test (10 Marks) and Innovative experiment/Concept Design and Implementation (10 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE REDUCED TO 30 MARKS.	30
MAXIMUM MARKS FOR THE CIE		100

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	CONTENTS	MARKS
PART A		
1	Objective type questions covering entire syllabus	10
PART B (Maximum of THREE Sub-divisions only)		
2	Unit - I: Compulsory	18
3 & 4	Unit - II: Question 3 or 4	18
5 & 6	Unit - III: Question 5 or 6	18
7 & 9	Unit - IV: Question 7 or 8	18
9 & 10	Unit - V: Question 9 or 10	18
TOTAL		100

RUBRICS FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	CONTENTS	MARKS
1	Write Up	10
2	Conduction of the Experiments	20
3	Viva	20
TOTAL		50

Syllabus class

Semester-related commands

Semester-related commands: Course definition and course type mapping



Head Of the Department (HOD)

- Defines the course code and course title.
- Maps the course to existing course type (defined by DA).
- Defines the Category and Assigns Board of Studies for the courses offered to the different departments.
- Group courses either as Basket Course group, Professional Elective group or Ability Enhancement group, apart from standalone courses.
- Verifies the total semester credit after the inclusion of courses.
- Generate <course-code>.tex files and assigns to subject experts

Semester-related commands: Category

Map **course types** defined under *Credit-related commands* to courses with

- Course-code with Course-title
- Category (shown to the left)
- BoS, Common to and Stream

General syntax:

```
\department[short form]{long form}  
\schemeYear[UG/PG]{Year}
```

```
\begin{semester}{semester number}  
  \begin{definecourse}{course code}  
    \category{text}  
    \stream{text}  
    \commonto{text}  
    \bos{text}  
    \coursetitle[course type]{title}  
  \end{definecourse}  
\end{semester}
```

Category:

- AC: Applied Science Course
- AE: Ability Enhancement Course[†]
- BC: Bridge Course
- ES: Emerging Science Course
- ET: Emerging Technology Course
- HS: Humanities and Social Sciences
- IE: Institutional Elective
- PC: Professional Core Course[†]
- PE: Professional Core Elective[†]
- PL: Programming Language Course
- SL: Skill Lab

[†]These category courses have special wrappers

Semester-related commands: Example of Bridge Course (BC) Category

8	MAT149AT	Bridge Course Mathematics	MA	3	0	0	0 [†]	Theory	2	100	—	—	100	—
9	SLCSP31A	Domain Engineering Skill Lab	CS	0	0	1	1	Lab	2	—	50	3	—	50
Total							23							

[†]Audit

Semester - III						
Bridge Course Mathematics						
Category: Bridge Course						
(Common to Common to CS,CD,CY and CI Programs)						
(Theory)						
Course Code	:	MAT149AT		CIE	:	100 Marks
Credit distribution: L:T:P	:	3:0:0		SEE	:	— Marks
Pedagogical Hours (L:T:P:S)	:	45:0:0:45		CIE Duration	:	2 Hours
Total credits:	:	3		SEE Duration	:	— Hours

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE SUM OF TWO WILL BE THE FINAL QUIZ MARKS.	20
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	40
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING	40

```

\begin{definecredits}
\begin{credit}{L0}◀
  \Lcredit{3}
  \CIEduration{2}
  \CIELmarks{100}◀
  \begin{CIerubrics}◀
    \Crubric{1}{\textbf{QUIZ...}}{20}
    \Crubric{2}{\textbf{TEST...}}{40}
    \Crubric{3}{\textbf{EXPE...}}{40}
  \end{CIerubrics}
\end{credit}
\end{definecredits}

```

Semester-related commands: Grouping courses

Supports 3 types of grouping:

1. Basket course group (using `coregroup` environment)
2. Ability Enhancement (AE) group (using `aegroup` environment)
3. Professional Elective (PE) group (using `electivegroup` environment)

Semester-related commands → category: Basket course grouping (PC)

```
\begin{semester}{semester number}
\begin{coregroup}◀
  \begin{definecourse}{course code}
    \category{PC}◀
    \stream{text}
    \commonto{text}
    \bos{text}
    \coursetitle[course type]{title}
  \end{definecourse}
  ...
  \begin{definecourse}{course code}
    \category{PC}◀
    \stream{text}
    \commonto{text}
    \bos{text}
    \coursetitle[course type]{title}
  \end{definecourse}
\end{coregroup}◀
\end{semester}
```

Syntax:

<code>\begin{coregroup}</code>	<code>\begin{coregroup*}</code>
<code>content...</code>	<code>content...</code>
<code>\end{coregroup}</code>	<code>\end{coregroup*}</code>

- Helps in creating Basket course group which accepts *only* **PC** based category courses.
- These courses will create entries in Index page (1st page) as well as in Credit page (2nd page) with \ as a separator between course-titles, course-codes and page numbers.
- This group doesn't generate any extra index page.

Example : Basket course grouping (PC)

```
\begin{semester}{3}
\begin{coregroup}
\begin{definecourse}{ea131aiX}
\category{PC}
\stream{EC}
\commonto{Common to EC, EI, EE Programs}
\bos{ECE}
\coursetitle[LOa]{Elements of electronics}
\end{definecourse}
\begin{definecourse}{eb131aiX}
\category{PC}
\stream{EC}
\commonto{Common to EC, EI, EE Programs}
\bos{ECE}
\coursetitle[LT]{Elements of electronics}
\end{definecourse}
\end{coregroup}
\end{semester}
```

Semester - III			
Sl. No.	Course Code	Course Title	Page No.
1	MA231TC	Linear Algebra and Probability Theory	1
2	EA131AIX / EB131AIX	Elements of electronics / Elements of electronics	2 / 4
3	EC132AXX	AE Course	6
4	EC134EXX	Advanced Electronics Group	10
5	EC131AIX	Elements of electronics	14

III SEMESTER														
Sl. No.	Course Code	Course Title	BoS	Credit Allocation			Category	CIE Duration (Hrs)	Max Marks CIE		SEE Duration (Hrs)	Max Marks SEE		
				L	T	P			Total	Theory		Lab	Theory	Lab
1	MA231TC	Linear Algebra and Probability Theory	MA	3	0	0	0 ¹	Theory	3	100	—	—	100	—
2	EA131AIX / EB131AIX	Elements of electronics	ECE / ECE	3	0	0	3	Theory	3	100	—	3	100	—
		Elements of electronics												
3	EC132AXX	AE Course	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
4	EC134EXX	Advanced Electronics Group	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
5	EC131AIX	Elements of electronics	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
				Total			15							

Note: The coregroup environment automatically uses the L:T:P distribution of first definecourse. In the example shown above, LOa credit is assigned to L:T:P, and the LT is not considered in the credit table.

Example: Basket course grouping (PC): Index Page

```
\begin{semester}{4}
\begin{coregroup*}◀
  \begin{definecourse}{ea131aiX}◀
    \category{PC}
    \stream{EC}
    \commonto{Common to EC, EI, EE Programs}
    \bos{ECE}
    \coursetitle[L0a]{Elements of electronics}◀
  \end{definecourse}◀
  \begin{definecourse}{eb131aiX}◀
    \category{PC}
    \stream{EC}
    \commonto{Common to EC, EI, EE Programs}
    \bos{ECE}
    \coursetitle[LT]{Elements of electronics}◀
  \end{definecourse}◀
\end{coregroup*}◀
\end{semester}
```

Semester - III			
Sl. No.	Course Code	Course Title	Page No.
1	MA231TC	Linear Algebra and Probability Theory	1
2	EA131AIX / EB131AIX	Elements of electronics / Elements of electronics	2 / 4
3	EC132AXX	AE Course	6
4	EC134EXX	Advanced Electronics Group	10
5	EC131AIX	Elements of electronics	14

Semester - IV			
Sl. No.	Course Code	Course Title	Page No.
1	EE141AIX	Elements of electronics	16
2	EA131AIX / EB131AIX	Elements of electronics / Elements of electronics	2 / 4
3	AY141AIX	Yoga	18

Example: Basket course grouping (PC): Credit page

```
\begin{semester}{4}
\begin{coregroup*}
\begin{definecourse}{ea131aiX}
\category{PC}
\stream{EC}
\commonto{Common to EC, EI, EE Programs}
\bos{ECE}
\coursetitle[L0a]{Elements of electronics}
\end{definecourse}
\begin{definecourse}{eb131aiX}
\category{PC}
\stream{EC}
\commonto{Common to EC, EI, EE Programs}
\bos{ECE}
\coursetitle[LT]{Elements of electronics}
\end{definecourse}
\end{coregroup*}
\end{semester}
```

III SEMESTER														
Sl. No.	Course Code	Course Title	BoS	Credit Allocation			Category	CIE Duration (Hrs)	Max Marks CIE		SEE Duration (Hrs)	Max Marks SEE		
				L	T	P			Total	Theory		Lab	Theory	Lab
1	MA231TC	Linear Algebra and Probability Theory	MA	3	0	0	0 ^a	Theory	2	100	—	—	100	—
2	EA131AIX / EB131AIX	Elements of electronics / Elements of electronics	ECE / ECE	3	0	0	3	Theory	2	100	—	3	100	—
3	EC132AXX	AE Course	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
4	EC134EXX	Advanced Electronics Group	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
5	EC131AIX	Elements of electronics	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
Total				15										

IV SEMESTER															
Sl. No.	Course Code	Course Title	BoS	Credit Allocation			Category	CIE Duration (Hrs)		Max Marks CIE		SEE Duration (Hrs)		Max Marks SEE	
				L	T	P		Theory	Lab	Theory	Lab	Theory	Lab		
1	EE141AIX	Elements of electronics	ME	3	1	0	4	Theory	2	100	—	3	100	—	—
2	EA131AIX / EB131AIX	Elements of electronics / Elements of electronics	ECE / ECE	3	0	0	3	Theory	2	100	—	3	100	—	—
3	AY141AIX	Yoga	XYZ	0	0	1	1	Lab	2	—	50	3	—	50	—
Total				8											

^aAudit

Semester-related commands → category: Ability Enhancement courses

Syntax:

```
\begin{semester}{semester number}
\begin{aegroup}{common code}◀
\begin{definecourse}{course code}
\category{AE}◀
\stream{text}
\commonto{text}
\bos{text}
\coursetitle[course type]{title}
\end{definecourse}
...
\begin{definecourse}{course code}
\category{AE}◀
\stream{text}
\commonto{text}
\bos{text}
\coursetitle[course type]{title}
\end{definecourse}
\end{aegroup}◀
\end{semester}
```

```
\begin{aegroup}{common code}
content...
\end{aegroup}
```

- Helps in creating AE course group which accepts *only* **AE** based category courses.
- These courses will create entries in Index page (1st page) as well as in Credit page (2nd page) with a course titled as AE course automatically with the course-code presented as common code (1st argument of aegroup environment).
- This group does generate an extra index page grouped under title **AE courses** with course code, course title and page numbers of individual courses.

Example : AE course grouping

```

\begin{semester}{3}
\begin{aegroup}{ec132aXX}
\begin{definecourse}{ec132a01}
\category{AE}
\stream{EC}
\commonto{Common to EC, EI, EE Programs}
\bos{ECE}
\coursetitle[LP]{Elements of electronics}
\end{definecourse}
\begin{definecourse}{ec132a02}
\category{AE}
\stream{EC}
\commonto{Common to EC, EI, EE Programs}
\bos{ECE}
\coursetitle[LP]{Elements of electronics}
\end{definecourse}
\end{aegroup}
\end{semester}

```

Semester - III			
Sl. No.	Course Code	Course Title	Page No.
1	MA231TC	Linear Algebra and Probability Theory	1
2	EA131AIX / EB131AIX	Elements of electronics / Elements of electronics	2 / 4
3	EC132AXX	AE Course	6
4	EC134EXX	Advanced Electronics Group	10
5	EC131AIX	Elements of electronics	14

III SEMESTER														
Sl. No.	Course Code	Course Title	BoS	Credit Allocation			Category	CIE Duration (Hrs)	Max Marks CIE		SEE Duration (Hrs)	Max Marks SEE		
				L	T	P			Theory	Lab		Theory	Lab	
1	MA231TC	Linear Algebra and Probability Theory	MA	3	0	0	0 ¹	Theory	3	100	—	—	100	—
2	EA131AIX / EB131AIX	Elements of electronics / Elements of electronics	ECE / ECE	3	0	0	3	Theory	3	100	—	3	100	—
3	EC132AXX	AE Course	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
4	EC134EXX	Advanced Electronics Group	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
5	EC131AIX	Elements of electronics	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
Total				15										

Semester - III			
AE Courses			
Sl. No.	Course Code	Course Title	Page No.
1	EC132A01	Elements of electronics	6
2	EC132A02	Elements of electronics	8

Note: The aegroup environment automatically uses the L:T:P distribution of first definecourse. In the example shown above, both the credits have LP credit and hence it is assigned to L:T:P in the credit table.

Semester-related commands → category: Professional Elective courses

```
\begin{semester}{semester number}
  \begin{electivegroup}{common code}◀
    {common title}{group letter}
    \begin{definecourse}{course code}
      \category{PE}◀
      \stream{text}
      \commonto{text}
      \bos{text}
      \coursetitle[course type]{title}
    \end{definecourse}
    ...
    \begin{definecourse}{course code}
      \category{PE}◀
      \stream{text}
      \commonto{text}
      \bos{text}
      \coursetitle[course type]{title}
    \end{definecourse}
  \end{electivegroup}◀
\end{semester}
```

Syntax:

```
\begin{electivegroup}{common code}{common
  title}{group letter}
  content...
\end{electivegroup}
```

- Helps in creating PE course group which accepts *only* **PE** based category courses.
- These courses will create entries in Index page (1st page) as well as in Credit page (2nd page) with a course title given as common title (2nd argument of electivegroup environment) with the course-code presented as common code (1st argument).
- This group does generate an extra index page grouped under title **Group <group letter>** with course code, course title and page numbers of individual courses.

Example : PE course grouping

```
\begin{semester}{3}
\begin{electivegroup}{ec134eXX}
  {Advanced Electronics Group}{A}
\begin{definecourse}{ec132e01}
  \category{PE}
  \stream{EC}
  \common{Common to EC, EI, EE Programs}
  \bos{ECE}
  \coursetitle[LP]{Elements of electronics}
\end{definecourse}
\begin{definecourse}{ec132e02}
  \category{PE}
  \stream{EC}
  \common{Common to EC, EI, EE Programs}
  \bos{ECE}
  \coursetitle[LP]{Elements of electronics}
\end{definecourse}
\end{electivegroup}
\end{semester}
```

Semester - III			
Sl. No.	Course Code	Course Title	Page No.
1	MA231TC	Linear Algebra and Probability Theory	1
2	EA131AIX / EB131AIX	Elements of electronics / Elements of electronics	2 / 4
3	EC132AXX	AE Course	6
4	EC134EXX	Advanced Electronics Group	10
5	EC131AIX	Elements of electronics	14

III SEMESTER														
Sl. No.	Course Code	Course Title	BoS	Credit Allocation				Category	CIE Duration (Hrs)	Max Marks CIE		SEE Duration (Hrs)	Max Marks SEE	
				L	T	P	Total			Theory	Lab		Theory	Lab
1	MA231TC	Linear Algebra and Probability Theory	MA	3	0	0	0 ¹	Theory	3	100	—	—	100	—
2	EA131AIX / EB131AIX	Elements of electronics / Elements of electronics	ECE / ECE	3	0	0	3	Theory	3	100	—	3	100	—
3	EC132AXX	AE Course	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
4	EC134EXX	Advanced Electronics Group	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
5	EC131AIX	Elements of electronics	ECE	3	0	1	4	Theory+Lab	1.5	100	50	3	100	50
Total				15										

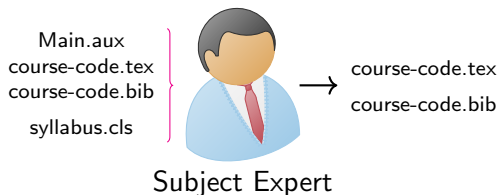
Semester - III			
Advanced Electronics Group			
GROUP-A			
Sl. No.	Course Code	Course Title	Page No.
1	EC132E01	Elements of electronics	10
2	EC132E02	Elements of electronics	12

Note: The aegroup environment automatically uses the L:T:P distribution of first definecourse. In the example shown above, both the credits have LP credit and hence it is assigned to L:T:P in the credit table.

Syllabus class

Course-related commands

Semester-related commands: Course definition and course type mapping



- Receive `<course-code>.tex` and `<course-code>.bib` files from HOD
- Defines the unit-wise lecture contents for course-types: LO, LT and LP.
- Defines the practical experiment list for course-types: PO and LP.
- Defines the list of course-outcomes.
- Use reference manager to build the references for a particular course and uses the citation keys to add references inside `<course-code>.tex` file.
- Return both the files back to HOD, which HOD uses to compile the syllabus book for that year students.

Course-related commands: Lecture Syntax

Subject experts defines

- Lecture units (in LO, LT, LP based course types)

```
\begin{units}  
  \unit[hours]{content}  
\end{units}
```

```
\begin{units}  
  \unit[8]{\textbf{BJT:}}\\  
  Review of BJT Amplifiers: The Common collector  
    amplifier, Emitter Follower, Darlington pair,  
    discrete amplifier design problems.\\  
  \textbf{MOS Field Effect Transistors (MOSFETs):}\\  
  Device structure and physical operation, current  
    voltage characteristics, Transfer  
    characteristics (both N-MOS and P-MOS), finite  
    output resistance in saturation. MOSFET circuits  
    at DC, Biasing in discrete MOS amplifier  
    circuits}  
  \unit[8]{\textbf{MOSFET amplifier:}}\\  
  Small signal operation and models: The DC bias point,  
    The signal current in the drain terminal, The  
    voltage gain, The small signal equivalent  
    circuit model, The trans conductance, body  
    effect, The common source amplifier (with  
    resistive load, diode connected load, current  
    source load).}  
\end{units}
```

Unit - I	8 Hours
BJT: Review of BJT Amplifiers: The Common collector amplifier, Emitter Follower, Darlington pair, discrete amplifier design problems. MOS Field Effect Transistors (MOSFETs): Device structure and physical operation, current voltage characteristics, Transfer characteristics (both N-MOS and P-MOS), finite output resistance in saturation. MOSFET circuits at DC, Biasing in discrete MOS amplifier circuits	
Unit - II	8 Hours
MOSFET amplifier: Small signal operation and models: The DC bias point, The signal current in the drain terminal, The voltage gain, The small signal equivalent circuit model, The trans conductance, body effect, The common source amplifier (with resistive load, diode connected load, current source load).	

Course-related commands: Practical Syntax

Subject experts defines

- Practical units (in LP and PO based course types)

```
\begin{practicals}  
  \practicetitle{title}  
  \practiceinstruction{instruction text}  
  \experiment{content}  
  \begin{subpracticals}  
    \subexperiment{content}  
  \end{subpracticals}  
  \experiment[tag]{content}  
\end{practicals}
```

Example in Next slide ...

Course-related commands: Practical Syntax (cont.)

```
\begin{practicals}
\practicetitle{Practicals: Hardware Experiments}
\practiceinstruction{Here you can put your
    instructions here. If you don't want this
    instruction, simply comment this line.}
\experiment{Characteristics of NMOS transistor (VI and
    Transfer)}
\experiment{Design and testing of RC phase shift
    oscillator.}
\experiment{Wien bridge oscillator circuits using
    operational amplifier.}
\experiment{Design \& testing of an RC coupled
    amplifier using BJT in CE configuration.}
\experiment{Design \& testing of LC Oscillators.}
\begin{subpracticals}
\subexperiment{Design and Performance Analysis of
    Hartley oscillator.}
\subexperiment{Design and Performance Analysis of
    Colpitts oscillator.}
\end{subpracticals}
\experiment{Design and testing of class B power
    amplifier circuits.}
\practiceeltitle{Innovative Experiments}
\experiment[EL]{R-2R Digital-to-Analog Converter.}
\experiment[EL]{Successive Approximation ADC.}
\end{practicals}
```

Practical's: Hardware Experiments

Here you can put your instructions here. If you don't want this instruction, simply comment this line.

1. Characteristics of NMOS transistor (VI and Transfer)
2. Design and testing of RC phase shift oscillator.
3. Wien bridge oscillator circuits using operational amplifier.
4. Design & testing of an RC coupled amplifier using BJT in CE configuration.
5. Design & testing of LC Oscillators.
 - a) Design and Performance Analysis of Hartley oscillator.
 - b) Design and Performance Analysis of Colpitts oscillator.
6. Design and testing of class B power amplifier circuits.

Innovative Experiments

7. [EL] R-2R Digital-to-Analog Converter.
8. [EL] Successive Approximation ADC.

Course-related commands: CO Syntax

Subject experts defines

- Course outcomes

```
\begin{courseoutcomes}  
  \co{content}  
\end{courseoutcomes}
```

```
\begin{courseoutcomes}  
  \co{Analyze the working of devices like MOSFETs and  
      OPAMPs.}  
  \co{Illustrate the working of precision rectifiers  
      and amplifiers.}  
  \co{Apply the knowledge to design amplifier,  
      precision rectifier and waveform generators.}  
  \co{Evaluate electronic sub systems with respect to  
      the desired specifications.}  
\end{courseoutcomes}
```

Course Outcomes: After completing the course, the students will be able to:
1. Analyze the working of devices like MOSFETs and OPAMPs.
2. Illustrate the working of precision rectifiers and amplifiers.
3. Apply the knowledge to design amplifier, precision rectifier and waveform generators.
4. Evaluate electronic sub systems with respect to the desired specifications.

Course-related commands: Reference Syntax

Subject experts defines

- Added references using citation keys

```
\begin{references}  
  \reference{bibkey}  
\end{references}
```

```
\begin{references}  
  \reference{Sedra2007}  
  \reference{Razavi2014}  
  \reference{Millman2010}  
  \reference{Boylestad2021}  
\end{references}
```

References

- [1] A. S. Sedra and K. C. Smith, *Microelectronic Circuits*. Oxford University Press, 2007, p. 1392, ISBN: 9780195338836
- [2] B. Razavi, *Fundamentals of microelectronics*, Second edition. Hoboken, NJ: Wiley, 2014, 903 pp., Includes index, ISBN: 1118156323
- [3] J. Millman, *Millman Electronic Devices and Circuits*. McGraw-hill, 2010, ISBN: 0070634556
- [4] R. L. Boylestad, *Electronic Devices and Circuit Theory (10th Edition)*. TBS, 2021, p. 892, ISBN: 9788131727003

How to create citation key, next slide ...

How to create citation-key

Creation of citation keys:

- Using JabRef - an open source reference manager, search under `ctrl + n` keystroke popup window:
 - Text books using ISBN number
 - Journals using DOI number
 - Youtube video:
 1. Select Misc from recommended option,
 2. Enter
Optional fields:: Title: *title of the video*, Author: *author name*, Note: *[Online; accessed YYYY-MM-DD]*
General:: url: *link*.
 3. Under Required fields, click on  Generate .
- Save the entries.
- Select each entry and copy the citation-key alone using `ctrl + shift + k` keystrokes.
- Paste the citation-key inside bibkey argument of `\reference{bibkey}` command.

Course-related commands: Editable CIE rubrics contents

Subject experts defines (w.r.t LP based course-type)

- Here subject experts based on the need can modify the content of CIE Rubrics alone.

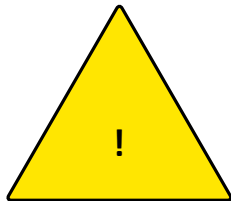
LP based course-type:

```
\CIErubricA{\textbf {QUIZZES:} Quizzes will be
conducted in online/offline mode. \textbf {TWO
QUIZZES} will be conducted and each Quiz will be
evaluated for 10 Marks. \textbf {THE AVERAGE OF
THE TWO WILL BE THE FINAL QUIZ MARKS.}}
\CIErubricB{\textbf {TESTS:} Students will be evaluated
in test, descriptive questions with different
complexity levels (Based on Revised Bloom's
Taxonomy Levels).\textbf {THREE} tests will be
con}
\CIErubricC{\textbf {EXPERIENTIAL LEARNING:} Students
will be evaluated for their creativity and
practical implementation of the problem. Case
study-based teaching learning (05), Program
specific requirements (05), Video based
seminar/presentation/demonstration (10), Tools
usage (20). \textbf {ADDING UPTO 30 MARKS.}}
\CIErubricD{\textbf {LAB:} Conduction of laboratory
exercises, lab report, observation and analysis
(30 Marks), Lab test (10 Marks) and Innovative
experiment/Concept Design and Implementation (10
Marks) adding up to 50 Marks. \textbf {THE FINAL
MARKS WILL BE REDUCED TO 30 MARKS.}}
```

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION		
#	COMPONENTS	MARKS
1	QUIZZES: Quizzes will be conducted in online/offline mode. TWO QUIZZES will be conducted and each Quiz will be evaluated for 10 Marks. THE AVERAGE OF THE TWO WILL BE THE FINAL QUIZ MARKS.	10
2	TESTS: Students will be evaluated in test, descriptive questions with different complexity levels (Based on Revised Bloom's Taxonomy Levels). THREE tests will be con	30
3	EXPERIENTIAL LEARNING: Students will be evaluated for their creativity and practical implementation of the problem. Case study-based teaching learning (05), Program specific requirements (05), Video based seminar/presentation/demonstration (10), Tools usage (20). ADDING UPTO 30 MARKS.	30
4	LAB: Conduction of laboratory exercises, lab report, observation and analysis (30 Marks), Lab test (10 Marks) and Innovative experiment/Concept Design and Implementation (10 Marks) adding up to 50 Marks. THE FINAL MARKS WILL BE REDUCED TO 30 MARKS.	30
MAXIMUM MARKS FOR THE CIE		100

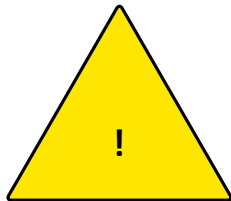
Few Cautions

Cautions w.r.t File usage



- Don't every change the name of `Main.tex` file.
- Before distributing `<course-code>.tex/.bib` files to Subject Experts, any change in Credits, Rubrics or Course order list will impact the compilation with Errors.
 - Sol.: Better delete All the `<course-code>.tex` as well as `<course-code>.bib` generated files.
 - Delete `Main.aux` file.
 - Recompile the `Main.tex` file again and share `<course-code>.tex + <course-code>.bib + Main.aux`.

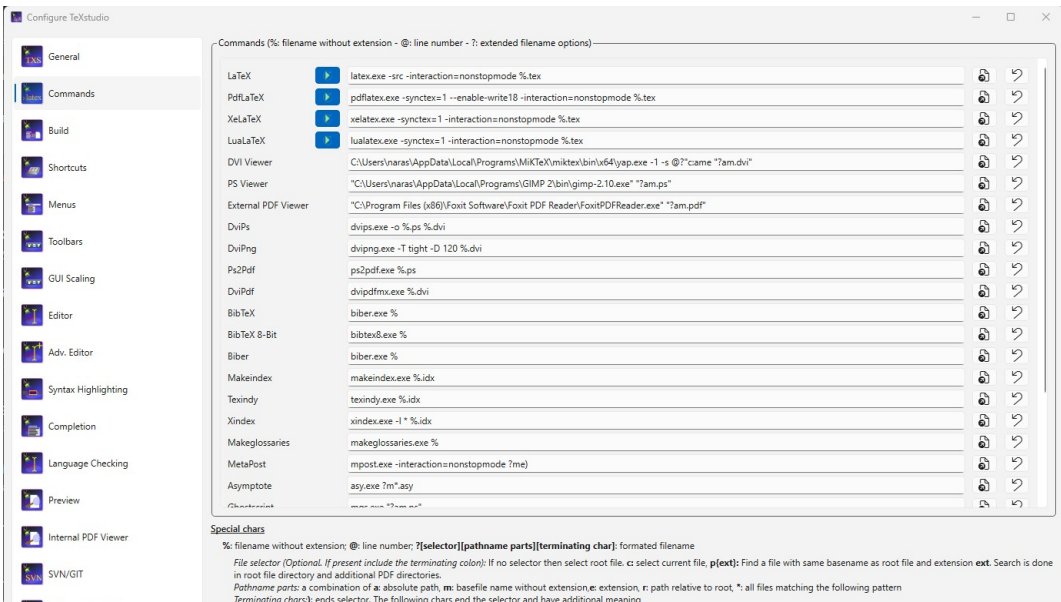
Cautions w.r.t contents



- Don't use & symbol directly in the text contents, you can use it as \&.
- Don't directly copy paste the contents, it may have unicode.
 - Sol.: Google search for "unicode online detector".
 - Replace the characters and paste the final text.

TexStudio Settings

TeXStudio Configuration setup



The screenshot shows the 'Configure TeXStudio' dialog box with the 'Commands' tab selected. The left sidebar lists various configuration categories, and the main area displays a list of commands with their corresponding paths and options. At the bottom, there is a section for 'Special chars' explaining the syntax for file selectors and pathnames.

Configure TeXStudio

Commands (%: filename without extension - @: line number - ?: extended filename options)

Command	Path/Options	Reset	Apply
LaTeX	latex.exe -src -interaction=nonstopmode %.tex		
PdLaTeX	pdflatex.exe -synctex=1 --enable-write18 -interaction=nonstopmode %.tex		
XeLaTeX	xelatex.exe -synctex=1 -interaction=nonstopmode %.tex		
LuaLaTeX	lualatex.exe -synctex=1 -interaction=nonstopmode %.tex		
DVI Viewer	C:\Users\naras\AppData\Local\Programs\MiKTeX\miktex\bin\x64\yap.exe -1 -s @"?*.dvi"		
PS Viewer	"C:\Users\naras\AppData\Local\Programs\GIMP 2\bin\gimp-2.10.exe" "%am.ps"		
External PDF Viewer	"C:\Program Files (x86)\Foxit Software\Foxit PDF Reader\FoxitPDFReader.exe" "%am.pdf"		
DviPs	dvips.exe -o %.ps %.dvi		
DviPng	dvipng.exe -T tight -D 120 %.dvi		
Ps2Pdf	ps2pdf.exe %.ps		
DviPdf	dvipdfmx.exe %.dvi		
BibTeX	biber.exe %		
BibTeX 8-Bit	bibtex8.exe %		
Biber	biber.exe %		
Makeindex	makeindex.exe %.idx		
Texindy	texindy.exe %.idx		
Xindex	xindex.exe -l * %.idx		
Makeglossaries	makeglossaries.exe %		
MetaPost	mpost.exe -interaction=nonstopmode ?me)		
Asymptote	asy.exe ?m*.asy		
Ghostscript	gs.exe "%am.ps"		

Special chars

?: filename without extension; @: line number; ?[selector][pathname parts][terminating char]: formatted filename

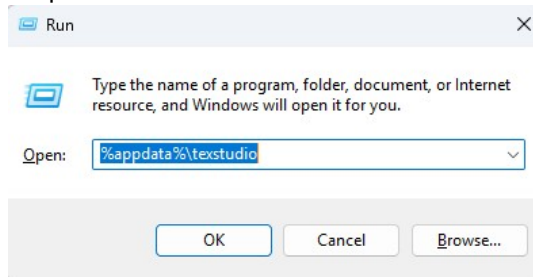
File selector (Optional. If present include the terminating colon): If no selector then select root file. **c**: select current file, **p(ext)**: Find a file with same basename as root file and extension **ext**. Search is done in root file directory and additional PDF directories.

Pathname parts: a combination of **a**: absolute path, **m**: basefile name without extension, **e**: extension, **r**: path relative to root, *****: all files matching the following pattern

Terminating chars: ends selector. The following chars end the selector and have additional meaning

TexStudio Auto-completion setup

Step 1:



Step 2: Open the completion → user folder and paste syllabus.cwl file into it

Step 3: In TexStudio, click on Options → Configure TexStudio → Completion and search for Syllabus

Conclusion & Acknowledgements

Special thanks to:

Dr. N. Ramavenkateswaran, Assistant Professor, Dept. of ECE - For planning and guiding.

Dr Veena Gadad, Associate Professor, Dept. of CSE - For guiding and debugging.